Challenges in Building KM Systems– Conventional Vs. KM System Life Cycle (KMSLS) – Knowledge Creation and Knowledge Architecture – Nonaka's Model of Knowledge Creation and Transformation .Knowledge Architecture

CHALLENGES IN BUILDING KM SYSTEM (8 marks)

Some of major challenges faced by knowledge management function are as follows −

- **Security** − Accommodating the right level of security for knowledge management is key. Conscious information should be shielded from most users, while allowing easy access to those with the proper credentials.
- **Getting People Motivated** − Overpowering organizational culture challenges and developing a culture that embraces learning, sharing, changing, improving can't be done with technology.
- **Keeping up with Technology** − Regulating how knowledge should be dispensed, transferring it quickly, and effectively is a huge challenge. Constantly changing structures mean learning how to be smart, quick, agile and responsive − all things a KM tool must be able to finish.
- **Measuring Knowledge** − Knowledge is not something that can be easily quantified, and is far more complex because it is copied out of human relationships and experience. The focus should be on distributed purpose rather than results or efforts.
- **Overpowering Shared Leadership** − As a knowledge leader, the concerned person has the responsibility to collaborate with fellow colleagues, persuade them to share their knowledge base for the benefit of the organization.
- **Keeping Accurate Data** − It is also the basic function to keep basic data which is accurate and authentic in nature.

challenges(12 marks)

1. Getting and Keeping People Motivated

It can prove challenging to get everyone on the same frequency when it comes to storing and sharing data. People (specifically, devs) are usually pretty comfortable with how they are used to doing things, making the adaptation stage a hurdle of its own. Getting your team motivated to share their work and ideas in a shared space is imperative to the success of your community and productivity. An easy way to get people on the same page from the get-go is to embody a culture of learning, improving, sharing, and change.

2. Keeping up with Ever-Changing Technology

Releases and updates are continuing to get more and more frequent when it comes to the technology we are using. Although there are constant changes nowadays with the technologies we utilize, it does instill a sense of urgency within us to keep up. Just remember this, changes

and updates to technology are a pain at first, but they usually open doors for new opportunities and can do more things for us in return.

3. Measuring Knowledge Contribution

Knowledge is not something that can be easily quantified and is far more complex because it is derived out of human relationships and experience. When it comes to knowledge contribution, the focus should be on the shared purpose rather than results or efforts.

4. Security

It is key (in most technical efforts) the provide the right level of security for your data, and it's no different for knowledge management. Use permissions and levels of security to make sure that various information is shared in the correct channels and with the right people.

5. Keeping Shared Information up to Date and Accurate

Moderation of your content **is challenging,** there's no denying it. Valuable data generated by your users may need routine updating and verification. Keep your information up-to-date by setting up validation reminders for your moderators or for the original posters.

6. Interpreting Data Effectively

Information derived by one group may need to be mapped or standardized in order to be meaningful to someone else in the organization.

7. Ensuring Relevancy

Data must support and answer questions being asked by your users. To do this effectively it requires appropriate meta-data is readily available for reference. Data relevancy means avoiding overloading users with unnecessary data and ensuring that content is moderated efficiently.

8. Rewarding Active Users

Choosing how you're going to reward your users is tough, and if you get enough contribution that you don't worry about what it would be like without those users, you may have never had this idea cross your mind. A community is only as strong as its users, so do what you can to recognize them for their knowledge contributions. It will not only encourage them to continue their interaction with your community, but it will encourage new users to jump on board and join your community.

9. Determining Who Will Be Responsible for Managing Your Knowledge

Choosing a moderator, community manager, or developer relations specialist is complex. You don't want to give the responsibility of the site to the wrong person, but without having run a community in the past it may be unclear who owns what. Break your community's goals out and determine who is responsible for the success of each of those KPIs, from there you can delegate who will be in charge of managing various pieces of knowledge.

10. Overcoming Shared Leadership

An extension of the above challenge, sometimes giving multiple people community responsibility can create a level of tension because many people feel attached in a way that they need to give their approval for various things.

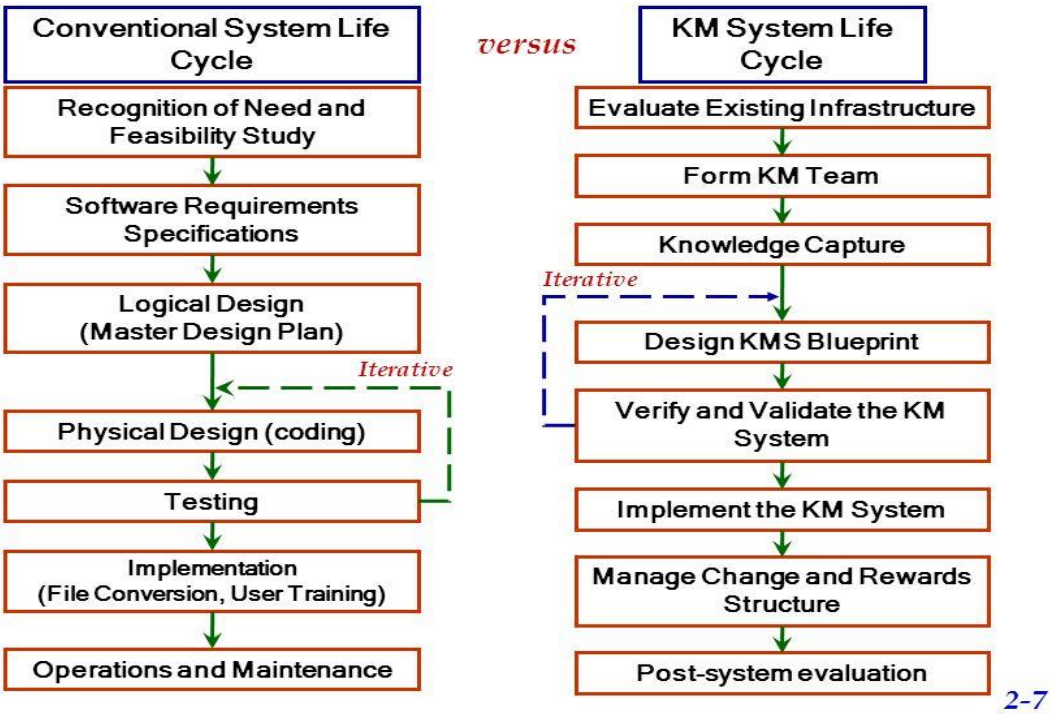
11. Turnover

There are two levels to this, and neither of them is very fun. Turnover is inevitable, and sometimes the responsibilities tied to your community can fall through the cracks. Keep a detailed log or manual of the tasks involved to keep your knowledge management system running smoothly. You can also experience a level of turnover at the user level. Some of your best users may have the time budgeted for you now, but what will you do manage all of their expertise once they move on. Create clear and easy to understand processes for data migration once people have left your company or no longer use your community.

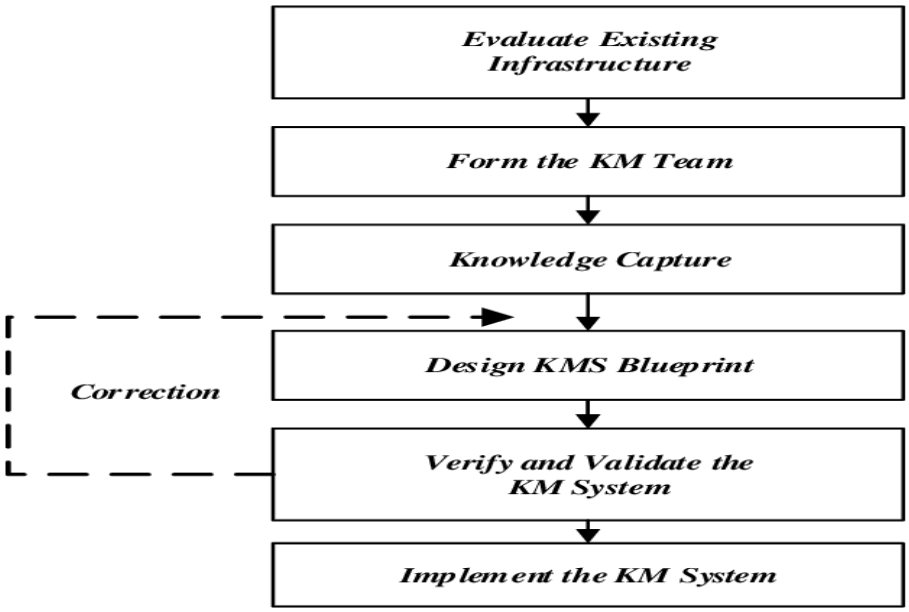
CONVENTIONAL Vs KMSLC

Key Differences

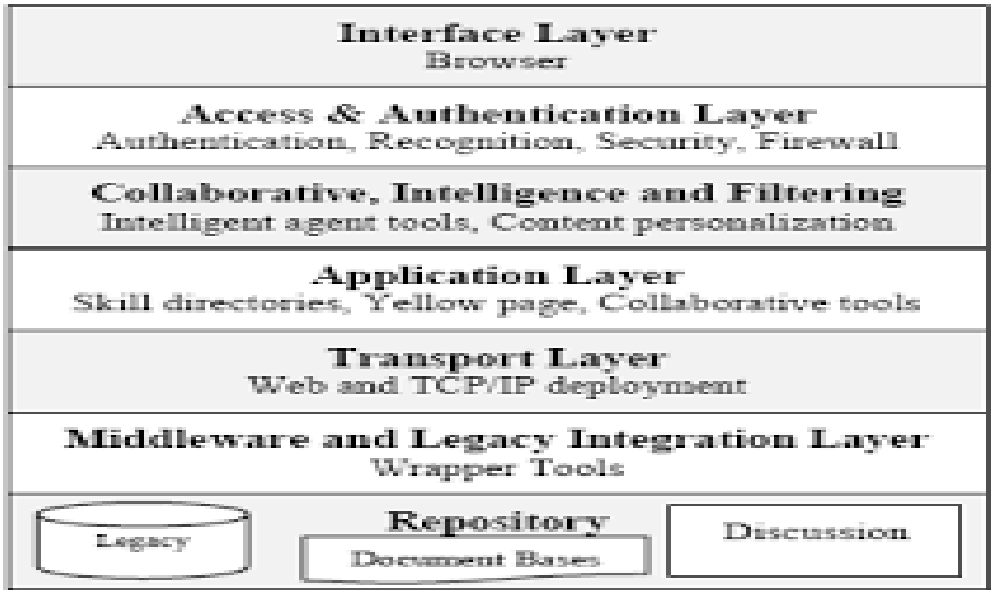
Conventional Systems	KM Systems
Systems analysts deal with information from the user	Knowledge developers deal with knowledge from domain experts
Users know the problem but not the solution	Domain experts know both the problem and the solution
System development is primarily sequential	System development is incremental and interactive
System testing normally at end of system life cycle	KM system testing evolves from beginning of the cycle

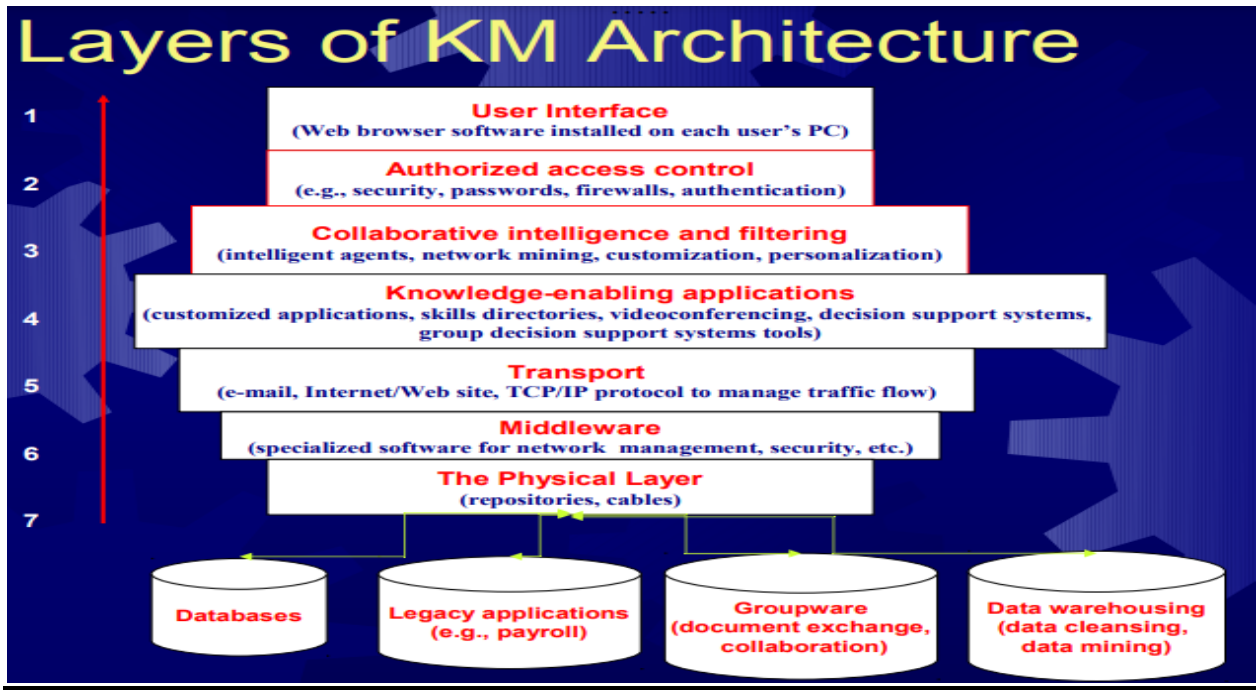


STEPS FOR BUILDING A KNOWLEDGE BASE



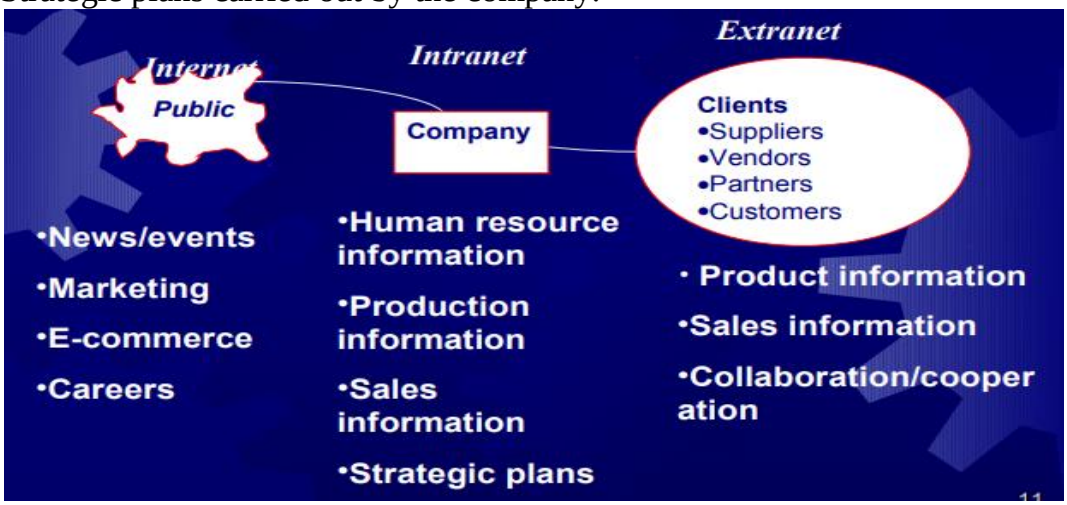
KNOWLEDGE CREATION AND KNOWLEDGE ARCHITECTURE





1. ***User Interface Layer***
   This layer focuses on consistency, relevancy, visual, clarity, navigation and usability**.**
2. ***Authorized Access Control***

It aims at maintaining controlled access of public over company data via internet such as Human resource information, production information, Sales information, Strategic plans carried out by the company.



3. ***Collaborative Intelligence and filtering***
    - ***Personalize*** *views based on stored knowledge.*
    - ***Groupware*** to facilitate both synchronous and asynchronous interaction and discussion.
    - ***Intelligent agents***- reduce search time for needed information.

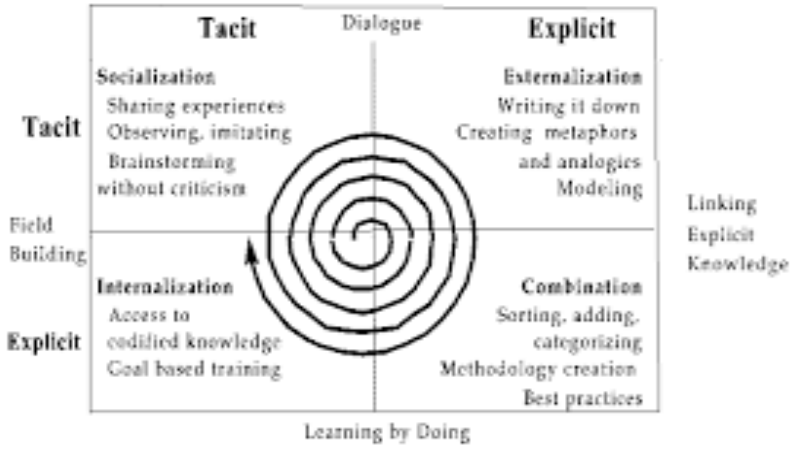
4. ***Knowledge Enabling application***
    - Also referred as Value-added layer.
    - Creates a competitive edge for the learning organization
    - Provides Knowledge bases, discussion databases, automation tools etc.
    - The ultimate goal is to show hoe knowledge sharing could improve the employees.

5. ***Transport Layer***
    - Most technical layer to implement.
    - Includes LANs, WANs, intranets, extranets, and the internet.
    - Ensures that the company will become a network of relationships.
    - It also considers multimedia, URLs, graphics, connectivity speeds and bandwidths.
6. ***Middleware***
    - Focus on interacting with legacy systems and programs residing on other platforms.
    - Designer should address databases and applications with which KM system interfaces.
    - Makes it possible to connect between old and new data.
7. ***Physical repositories***
    - It is the bottom layer in the KM architecture.

- Represents the physical layer where repositories are installed.
- Includes intelligent data warehouses, legacy applications, operational databases and special applications for security and traffic management.

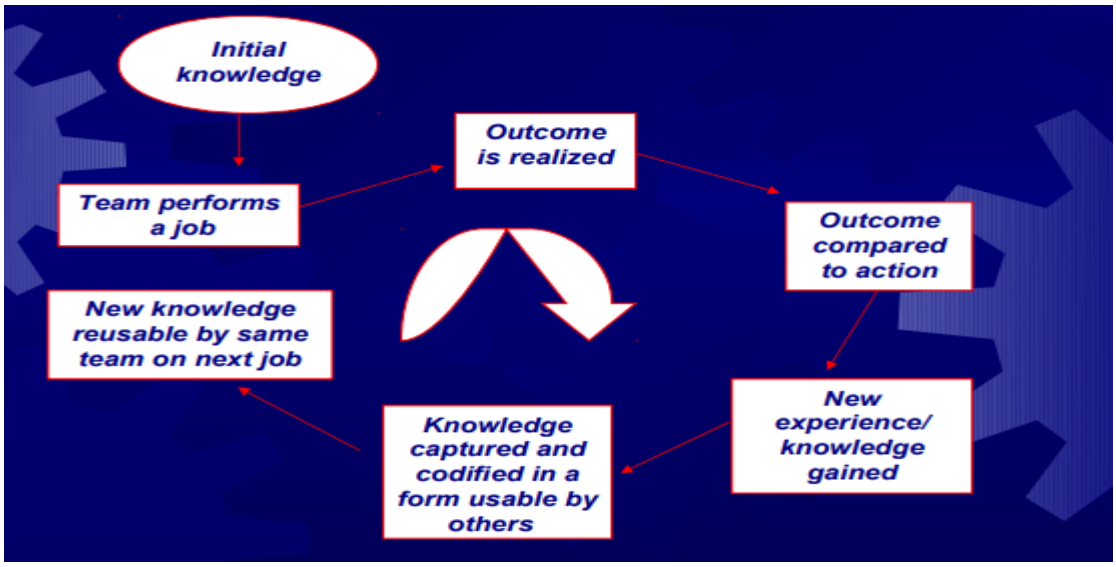
NONAKA TAKEUCHI MODEL

The Nonaka and Takeuchi model is one of the knowledge management model that can be used in the knowledge creation and conversion. Knowledge management models itself are the approach to manage knowledge from people and process of an organization. Each organization usually capture, codify, or render tangible and intangible knowledge in the shape that we know as tacit or explicit knowledge. Tacit knowledge is the kind of knowledge which is not written and used only by verbalizing the knowledge. This can lead to difficulties of acceptance for other people because there is no written information that can be used or read over in a long time, only the person who knows about the knowledge can keep it, but it will remain in the mind. In other words, explicit knowledge is the reverse explanation of tacit knowledge. Explicit knowledge is the written type of knowledge that can be used over by other people, but not all knowledge is transferred in a written way. Therefore, The Nonaka and Takeuchi model is one of the model that can represent the conversion of knowledge, whether it is from tacit to explicit or the other way around. Below is the figure of The Nonaka and Takeuchi model:



As we can see from the figure, there are four types of conversion of tacit and explicit knowledge. Below is the explanation of each types along with the example:

1. **Socialization**: it is the process of conversion from tacit to tacit knowledge. As we know, tacit is the unwritten type of knowledge. The most common example is training or sharing face to face. In training or sharing, the knowledge conversion is more natural because it is one of the typical social interactions. From doing this knowledge sharing or training, the knowledge will remain in the participant's mind, rather than noted elsewhere. This type is more to the process of acquiring it by experiencing the knowledge transmission.
2. **Externalization**: this type is conversing from tacit to explicit knowledge. From the unwritten knowledge, it is conversed to a written knowledge. The example is from someone who has a knowledge and starting to write a book to share the knowledge to many other people, so the knowledge can be tangible and permanent.
3. **Combination**: the next type is conversing from explicit to other explicit knowledge. This type can be interpreted to process of combining explicit knowledge form many sources and finally make a new form of explicit knowledge. The common example is when we are making a report or analysis. We search or gather explicit information about the transactions to make a financial report.
4. **Internalization**: it is the conversion from explicit to tacit knowledge. For this type, the knowledge is shaped in a written type first, then along with the time goes by, the knowledge will spread and become a tacit knowledge. An example for this is in an organization. Organization always has written rules for the company, but as the time goes by, the rules spread by the employee itself to the other employee and the rules become tacit knowledge because they already have the rules on their own mind.

KNOWLEDGE CREATION AND TRANSFORMATION

(Refer class notes- Knowledge sharing by team)